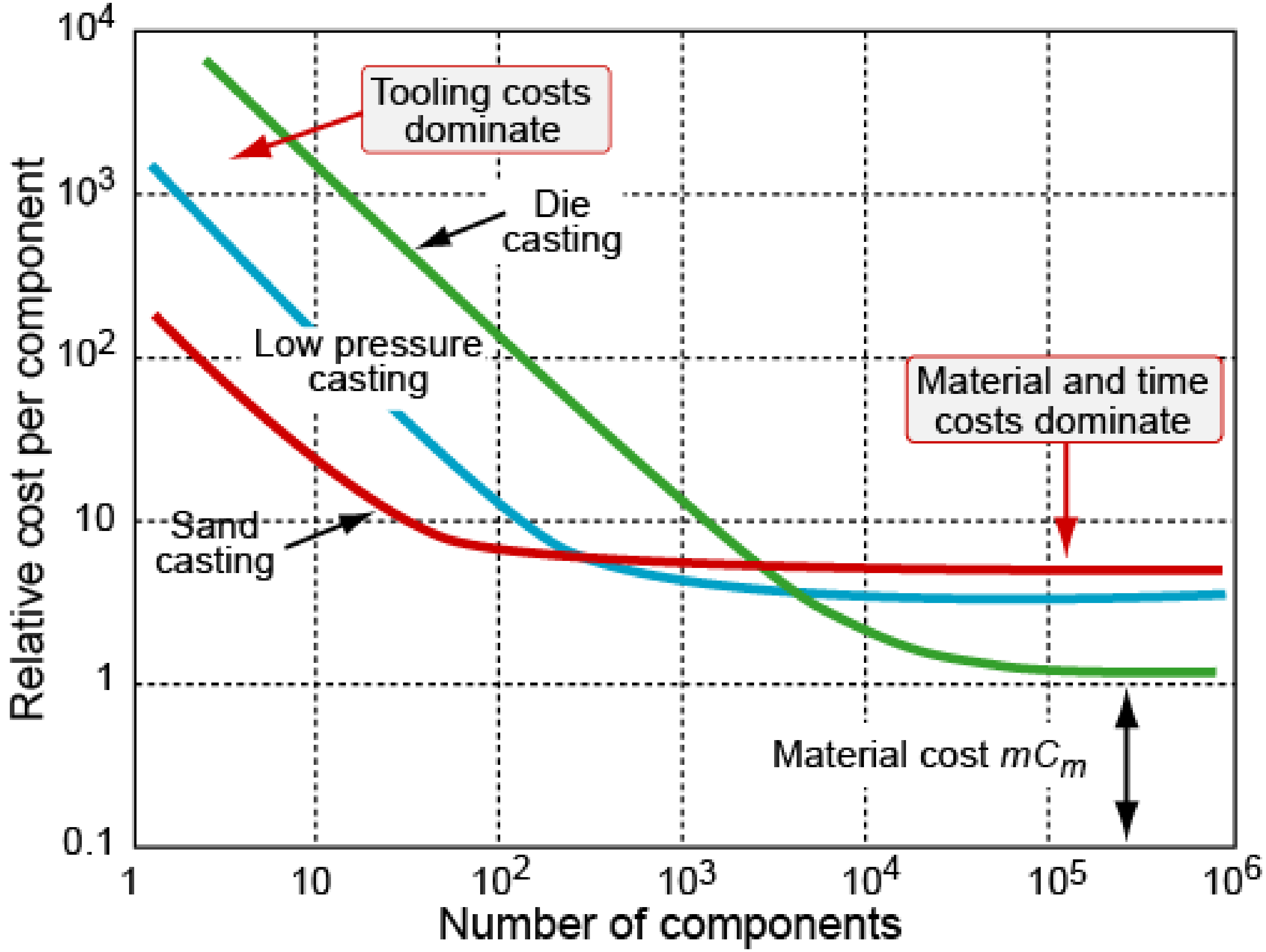
Lecture 13 - Cost Modelling in Design

James Busfield



Outline: Estimating Product Costs

- What will the customer afford?
- Purpose of cost estimates
- Inputs to a cost model
- The model and its implementation
- Cost drivers, batch size, assembly
- Resolving conflicting requirements

Resources:

“Materials: engineering, science, processing and design”, 2nd edition, Chapter 18

“Materials Selection in Mechanical Design”, 4th edition, Chapters 13 and 14

ANSYS Edupack Software

What Determines an Item’s Cost?

Consumers purchase goods and services to perform specific tasks.

The choice is based upon either price or performance or some intuitive ratio of the two.

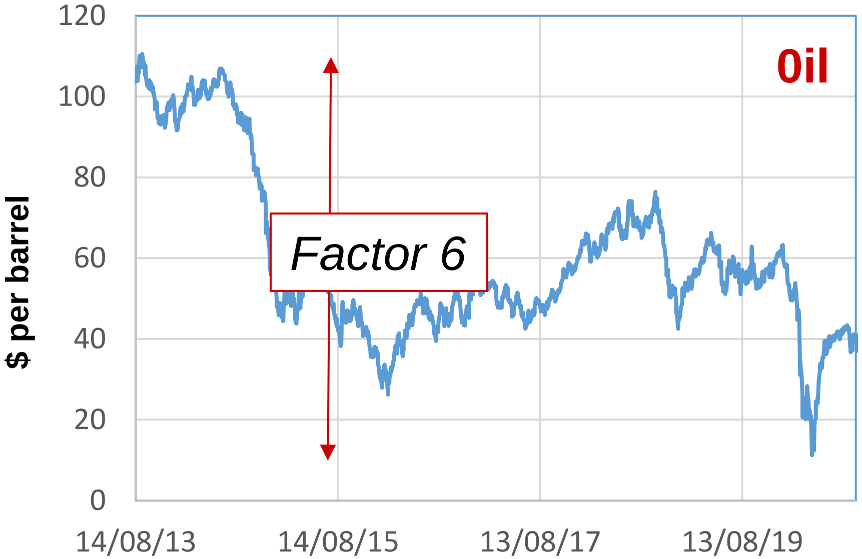
The real requirement is

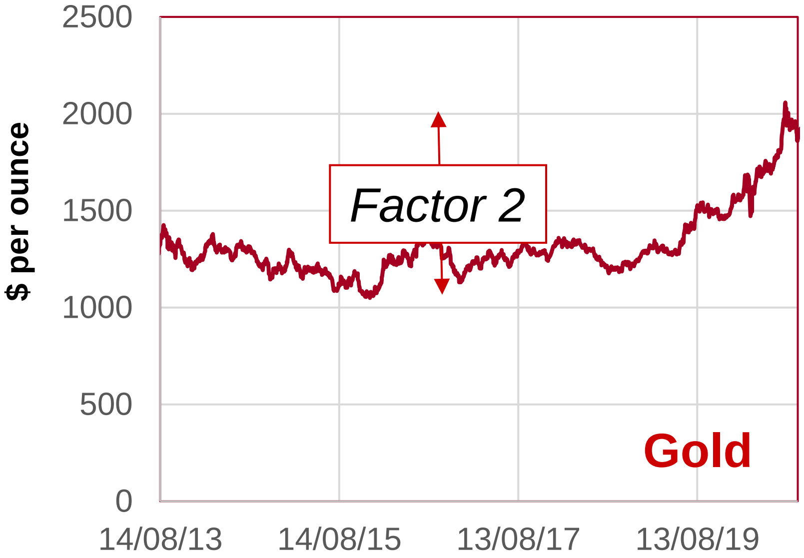
Cost < Price < Value

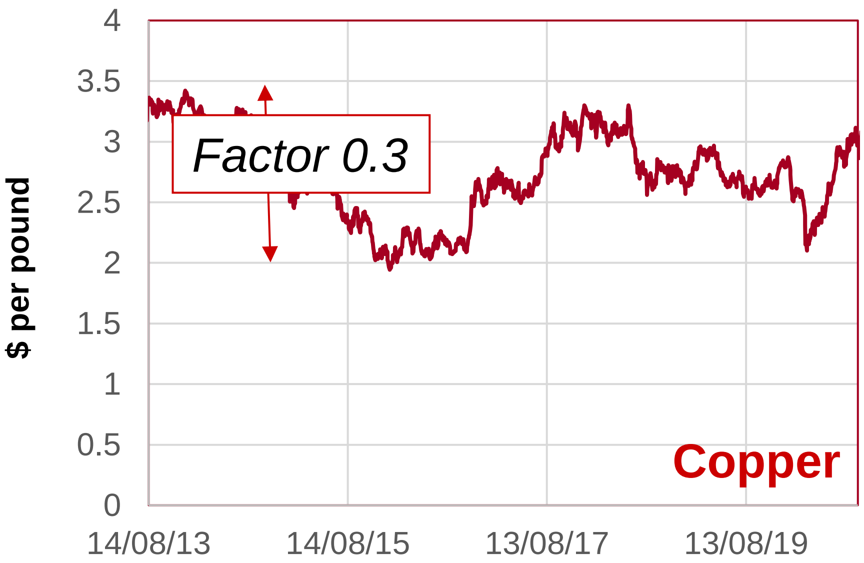
Cost =	the actual cost to make the part or product
Price =	the sum you sell it for
Value =	the worth the consumer puts on the product

Not worth the price	**= P > V**
Good value for money	**= P < V**

Raw Materials Price Volatility 2013-20







Cost modelling is essential in the design process for material selection

Cost of ‘Equivalent’ Stereo Systems

A stereo system has a wide range of cost and performance possibilities
The relative importance of these is subjective

QTX Amp - £55.24 (Amazon)

Bluetooth connectivity - Yes

Wireless connectivity - Yes

Built-in media player with USB and SD card inputs and FM receiver

A high output power of 55W max. per side

3.5mm stereo line input on the front panel,

2 stereo line inputs for RCA on the back,

2 x 6.3mm jack microphone inputs

Linn System Hub - £17,500 (Peter Tyson)

Main features	Connections for digital & analogue sources, wireless streaming, HDMI switching,
Audio formats	FLAC, ALAC, WAV, DSD (64/128/256), MP3, WMA (except lossless), AIFF, AAC, OGG
Resolution	Up to 24-bit 384 kHz
Integrated Services	Tidal, Qobuz, Spotify Connect, Airplay, TuneIn, Calm Radio
Optical Ethernet	Yes (SFP socket)
WiFi	Yes
Bluetooth	Yes (4.2)
Screen Type	1600 x 480 TFT display
Construction	Machined from Solid Aluminium
Finish	Black Anodised
Weight	16.4 kg

Other Examples

Citroen Ami

(£7,875)

Rolls-Royce Droptail

(£23,000,000)

Cost Estimation

Crude rule-of-thumb: when relatively low-tech products are mass produced in very large numbers, the product cost falls to 3 to 5 times the material cost. This is not very accurate though.

Cost estimate: competitive bidding -

An absolute cost is wanted, **± 3 %**

Very detailed inputs are needed

Cost estimate for initial selection -

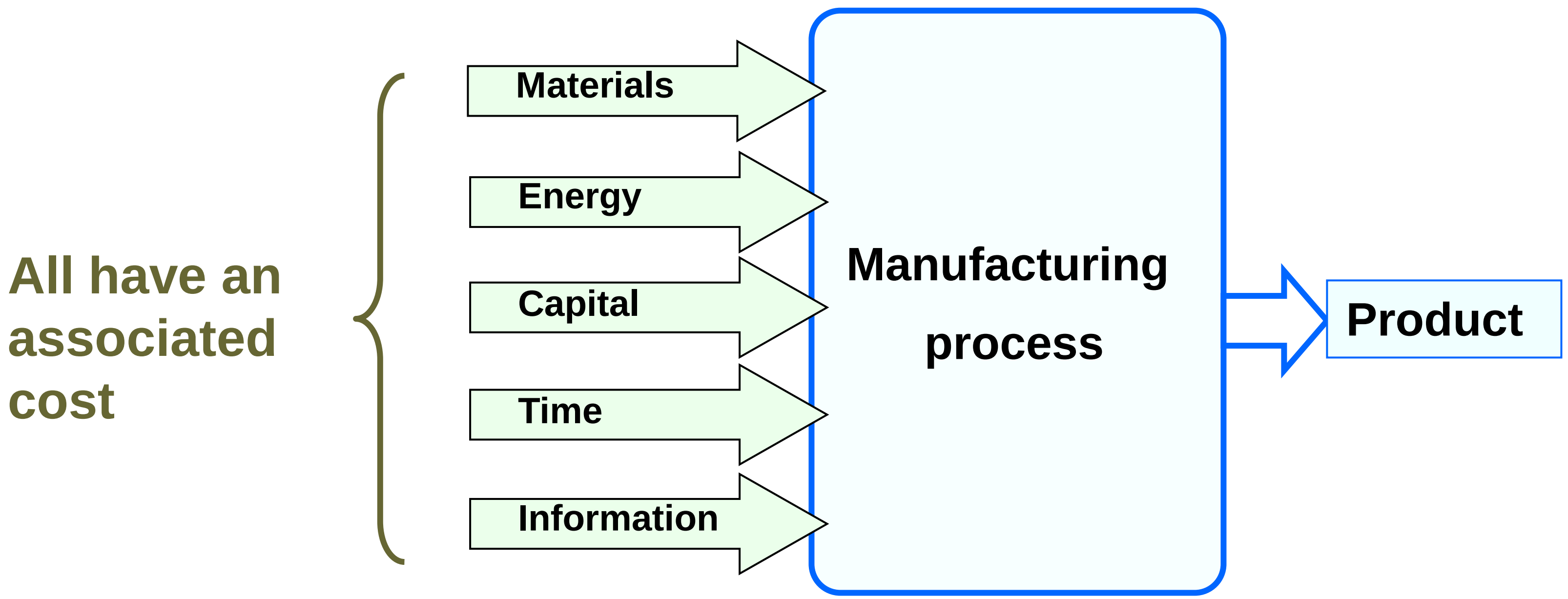
A relative cost is wanted, **± 30 %**

Much simpler inputs are OK

Estimating cost

When alternative material-process combinations meet the constraints, it is logical to rank them by **cost**

Generic inputs to any manufacturing process:



Inputs to a Cost Estimate

Resource		Symbol	Unit
Materials including consumables		C_{m}	£/kg
Capital	cost of equipment cost of tooling	C_{c} C_{t}	£ £
Time (including labour) overhead rate		$\dot{C}_{\mathrm{oh}}$	£/hr
Energy	power consumed cost of energy	P C_{e}	kW £/kW.hr
Space, **Admin.** a cost/hr		$\dot{C}_{\mathrm{s,a}}$	£/hr
Information	R & D royalties, licenses	$\dot{C}_{\mathrm{i}}$	£/hr

Lump into overhead rate $\dot{C}_{\mathrm{oh}}$

The Cost per Unit of Output

Material costs C_m, and a mass m is used per unit;
f is the scrap fraction (the fraction thrown away)

$$\frac{m C_m}{1-f}$$

Tooling C_t is "dedicated" -- it is written off against the number of parts to be made, n

$$\frac{C_t}{n}$$

Capital cost C_c of equipment is "non-dedicated" (it can be used for other products). It is written off against time, giving an hourly rate. The write-off time is t_{wo}. The rate of production is $\dot{n}$ units/hour. The load factor (fraction of time the equipment is used) is L.

$$\frac{1}{\dot{n}}\left(\frac{C_c}{L.t_{wo}}\right)$$

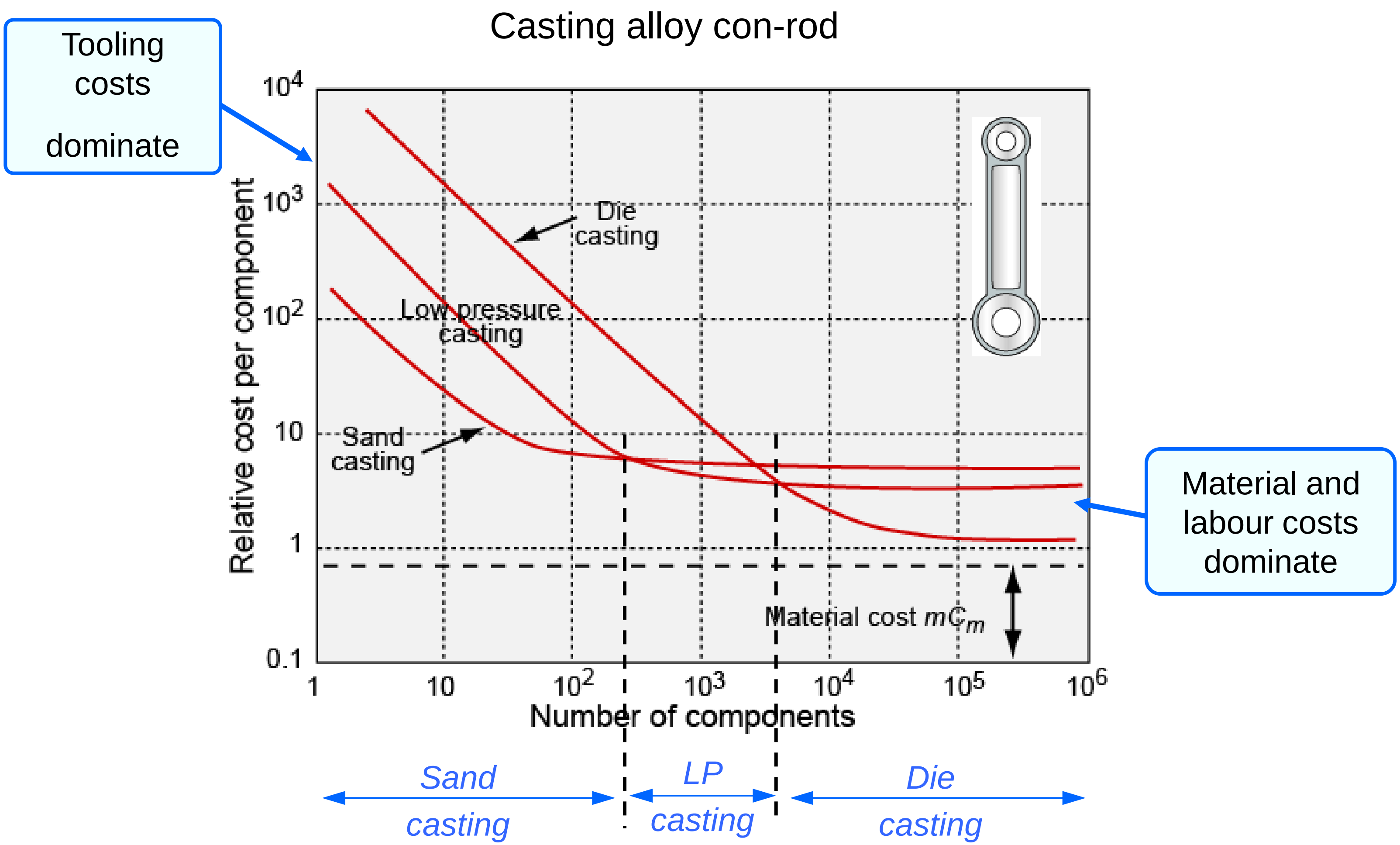
The gross **overhead** rate $\dot{C}_{oh}$ contributes a cost per unit of time that, like capital, depends on production rate $\dot{n}$

$$\frac{\dot{C}_{oh}}{\dot{n}}$$

$$C = \underbrace{\left[\frac{m\,C_m}{1-f}\right]}_{\text{Materials}} + \underbrace{\left[\frac{C_t}{n}\right]}_{\text{Tooling}} + \frac{1}{\dot{n}}\left[\frac{C_c}{L.t_{wo}} + \dot{C}_{oh}\right]$$

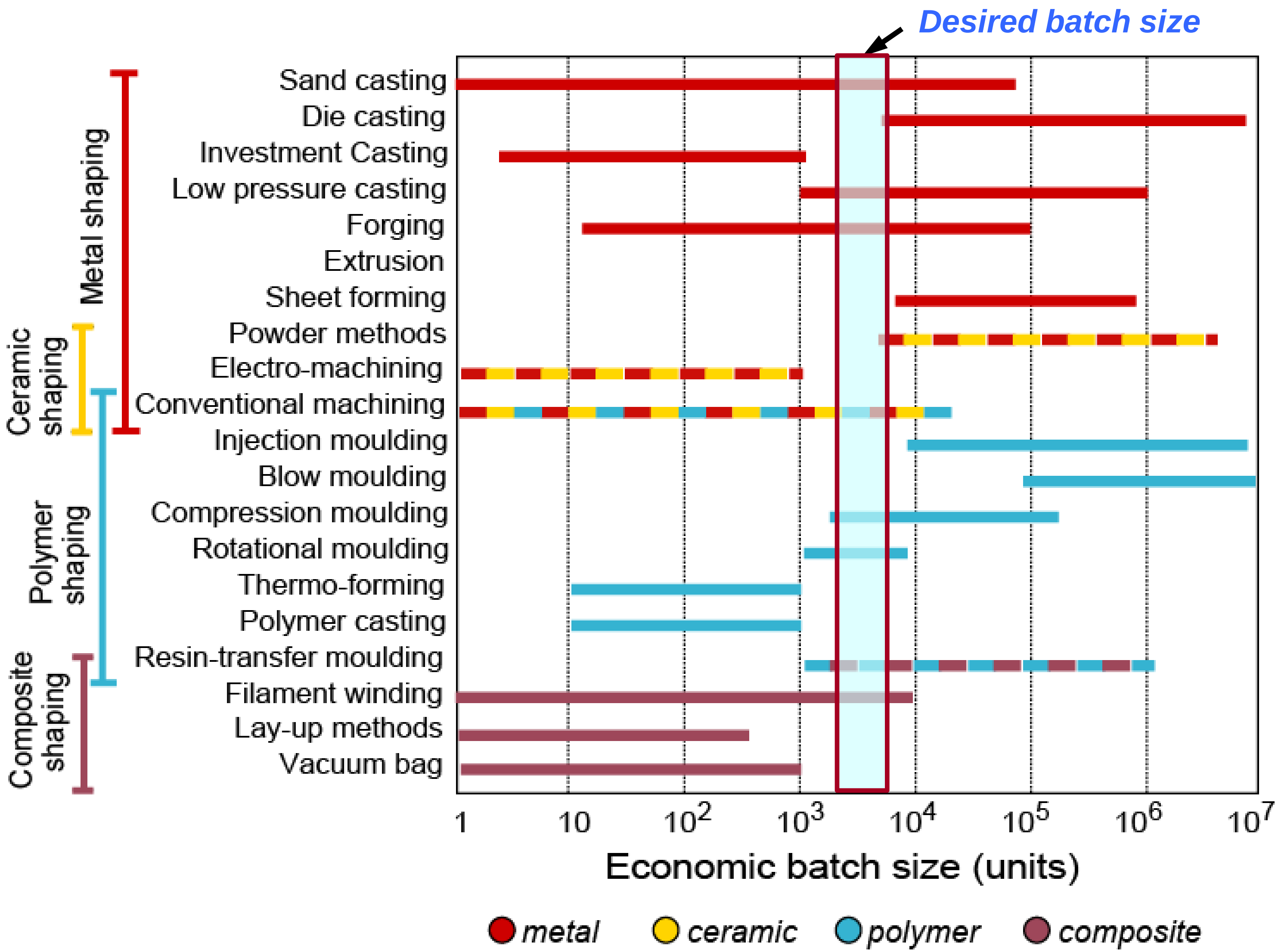
Materials; Tooling; Labor, Capital, Information, Energy...; Batch size (n); Rate of production ($\dot{n}$)

Features of a cost model



- Identify most economic process
- Examine materials-cost sensitivity
- Explore alternative materials and processes

Economic batch size



Where do you get the input information?

Queen Mary University of London

Material and process costs vary with time and depend on the quantity you order

CES EduPack has approximate cost for ~ 4,000 materials, which are all updated annually

The internet can help with commodity material prices for example:

- Iron & Steel Statistics Bureau - www.issb.co.uk
- Kitco Inc Gold & Precious Metal Prices - www.kitco.com

Ask suppliers: but how find them?

- Thomas Register of European Manufacturers – www.thomasnet.com

Cost modelling in CES

$$C = \left[\frac{m\,C_m}{1-f}\right] + \left[\frac{C_t}{n}\right] + \frac{1}{\dot{n}}\left[\frac{C_c}{L.t_{wo}} + \dot{C}_{oh}\right]$$

Characteristics of the process

Cost of equipment	C_c
Cost of tooling	C_t
Production rate	$\dot{n}$

The database has approximate value-ranges for these

Site-specific, user defined parameters

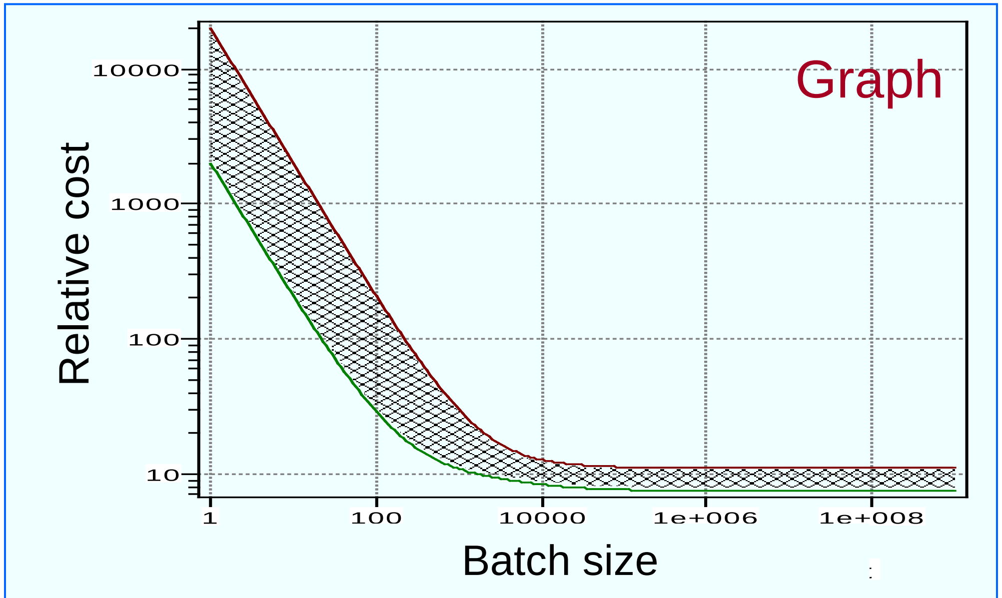
Batch size	n
Mass of component	m
Capital write-off time	t_{wo}
Load factor	L
Overhead rate	$\dot{C}_{oh}$

These are entered by the user via a dialog box

Cost model in CESEduPack Levels 2 and 3

Cost modeling

Relative cost index (per unit)	5	-	6	
Capital cost	2000	-	5000	GBP
Material utilisation factor	0.7	-	0.75	
Production rate (units)	20	-	30	per hr.
Tooling cost	300	-	450	GBP
Tooling life	5000	-	10000	units



Dialog box

Capital write-off time t_{wo} = ….

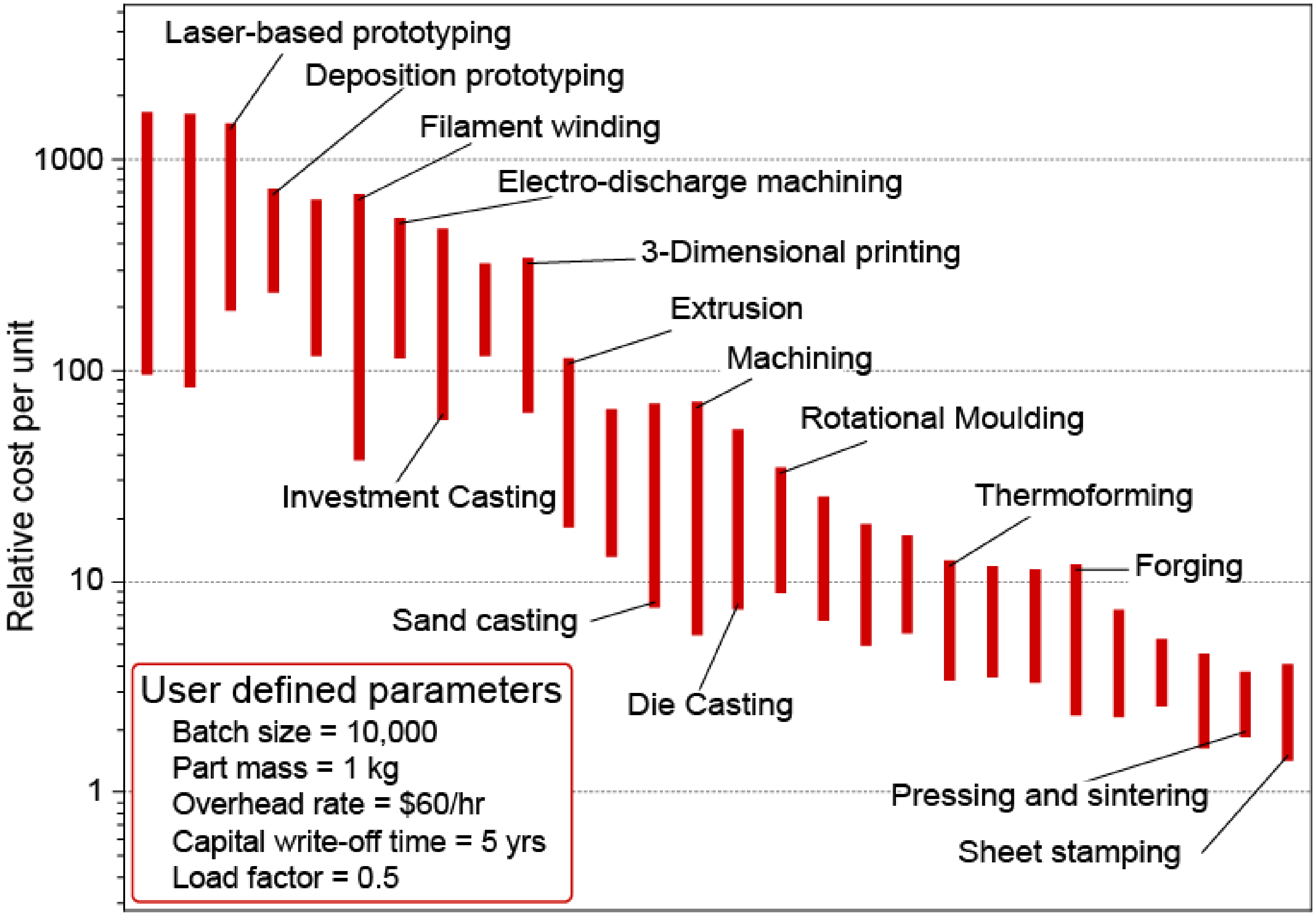
Component mass m = ….

Load factor L = ….

Material cost $\dot{C}_m$ =

Overhead rate C_{oh} = ….

Cost model in CES EduPack Level 3



Compulsory Exam Question from 20XX

- A 0.100kg polymer component can be manufactured by either machining from an extruded bar or by compression moulding from pellets.
- At this stage it is not known from your marketing department the precise production quantity, but you have been asked to calculate the cost per part of manufacturing 100, 1,000, 10,000, 100,000 and 1,000,000 by both processes ignoring the capital and labour costs whilst assuming the following information:
  - Polymer material cost - £2.00 /kg
  - The waste material created per part by machining - 0.100kg
  - The amount of waste material created per part by moulding - 0.010kg
  - The tooling costs dedicated for machining this part are - £10,000
  - The tooling costs dedicated for moulding this part are - £20,000

You should calculate the cost per part

- Which can for this example simplify the equation to just:

The cost per part	=	The Material cost per part	+	The Equipment cost amortised over production run

$$C = M'C_{\mathrm{m}} + \left[\frac{C_{\mathrm{t}}}{n}\right]$$

M' for machining = 0.20kg

M' for moulding = 0.11kg

C_{m} for both materials = £ 2.00/kg

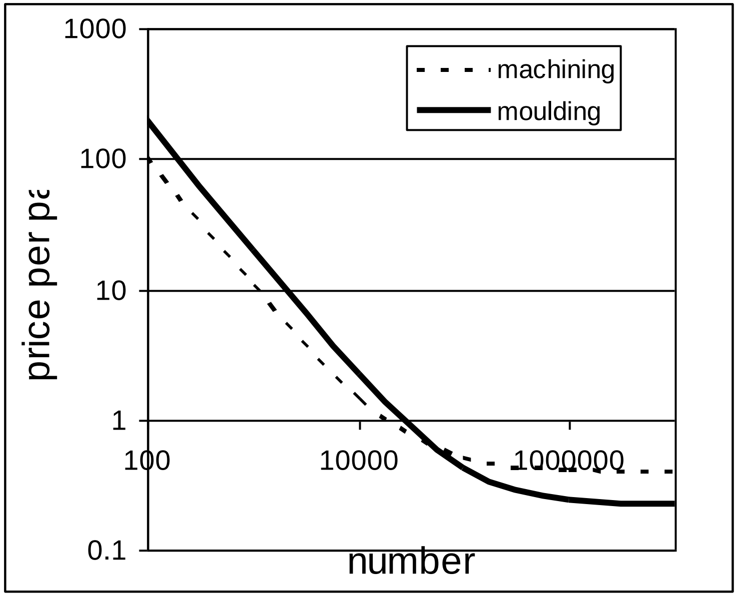
Calculate it for each process.

- For Machining

$$C = 0.20 \text{ x } £2.00 + \left[\frac{£10{,}000}{n}\right]$$

- For Moulding

$$C = 0.11 \text{ x } £2.00 + \left[\frac{£20{,}000}{n}\right]$$



n	Machining	Moulding
1	£10000.40	£20000.22
10	£1000.40	£2000.22
100	£100.40	£200.22
1000	£10.40	£20.22
10000	£1.40	£2.22
100000	£0.50	£0.42
1000000	£0.41	£0.24
10000000	£0.40	£0.22
55556	£0.58	£0.58

Joining: Design for Assembly

- Cost = sum of cost of the components + cost of assembly

Assembly is one of the bigger cost drivers

- Design for assembly addresses these key questions:

Is the part used only for fastening or connection? } If **YES**, is it needed?

Does it move relative to other parts?
Must it be a different material?
Must it be separated for disassembly or replacement?
} If **NO**, is it needed?

Can a change of shape-process integrate parts?
Can a change of joining-process allow faster assembly?
} If **YES**, consider implementation

Surface Treatment: Do NOT Overspecify

Over-specifying surface finish drives up cost without increasing value

- Functions achieved by a surface treatment:

Chemical:	corrosion and oxidation resistance
Mechanical:	hardness, wear, fatigue resistance
Thermal:	conduction and insulation
Electrical:	conduction and insulation
Aesthetic:	colour, reflectivity, texture

- Key questions:

  Is the treatment essential?

  If not essential for function, does it add perceived value?